Hussain Ahmad Madni

hamadnig@gmail.com Hussain Ahmad Madni Homepage hussain-ahmad

Research Interest

I am interested in addressing data and model heterogeneity challenges in cross-domain learning, with a focus on medical imaging and semantic segmentation. My research centers on advanced deep learning methods for medical image analysis, particularly segmentation and disease classification. Ultimately, I aim to develop robust AI-driven diagnostic tools to support clinical decision-making in healthcare.

Education

2022 – 2024 **PhD**, Computer Science and Artificial Intelligence, University of Udine, Italy.
Research Topic: *Data and Model Privacy in Cloud Computing using Federated Learning.*

2017 – 2018 **MS**, Computer Science, COMSATS University Islamabad, Pakistan.
Thesis title: *Intrusion Detection System Using Deep Learning.*

2010 – 2014 **BSc**, Computer System Engineering, The Islamia University of Bahawalpur, Pakistan.
Thesis title: *Targeted Malicious Email Detection using Machine Learning.*

2008 – 2010 **Intermediate (FSc Pre-Engineering)**, BISE, D. G. Khan, Pakistan.
Main Courses: *Mathematics, Physics, Chemistry, and English.*

2005 – 2007 **Matriculation (Science)**, BISE, D. G. Khan, Pakistan.
Main Courses: *Mathematics, Physics, Chemistry, Biology, and English.*

Professional Experience

2025 **Postdoctoral Researcher** University of Udine, Italy.

2019 – 2021 **Research Assistant** COMSATS University Islamabad, Pakistan.

2015 – 2019 **Software Engineer** Integrated Dynamic Solutions (IDS), Rawalpindi, Pakistan.

Research Publications

Journal Articles

1. **H. A. Madni**, H. Shujat, A. De Nardin, S. Zottin, and G. L. Foresti, "FsBAD: Data-efficient feature reconstruction for few-shot brain anomaly detection," *Pattern Recognition Letters*, 2025. DOI: 10.1016/j.patrec.2025.11.016.

2. **H. A. Madni**, H. Shujat, A. De Nardin, S. Zottin, and G. L. Foresti, "Unsupervised Brain MRI Anomaly Detection via Inter-Realization Channels," *International Journal of Neural Systems*, 2025. DOI: 10.1142/S0129065725500479.

3. **H. A. Madni**, R. M. Umer, S. Zottin, C. Marr, and G. L. Foresti, "FL-W3S: Cross-domain federated learning for weakly supervised semantic segmentation of white blood cells," *International Journal of Medical Informatics*, p. 105 806, 2025. DOI: 10.1016/j.ijmedinf.2025.105806.

4. **H. A. Madni**, R. M. Umer, and G. L. Foresti, "Exploiting data diversity in multi-domain federated learning," *Machine Learning: Science and Technology*, vol. 5, no. 2, p. 025 041, 2024. DOI: 10.1088/2632-2153/ad4768.

5. **H. A. Madni**, R. M. Umer, and G. L. Foresti, "Robust Federated Learning for Heterogeneous Model and Data," *International Journal of Neural Systems*, 2024. DOI: 10.1142/S0129065724500199.

6. **H. A. Madni**, R. M. Umer, and G. L. Foresti, "Blockchain-Based Swarm Learning for the Mitigation of Gradient Leakage in Federated Learning," *IEEE Access*, vol. 11, pp. 16 549–16 556, 2023. DOI: 10.1109/ACCESS.2023.3246126.

7 **H. A. Madni**, R. M. Umer, and G. L. Foresti, "Swarm-FHE: Fully Homomorphic Encryption based Swarm Learning for Malicious Clients," *International Journal of Neural Systems*, 2023. DOI: 10.1142/S0129065723500338.

8 F. Abdullah, R. Imtiaz, **H. A. Madni**, *et al.*, "A Review on Glaucoma Disease Detection Using Computerized Techniques," *IEEE Access*, vol. 9, pp. 37 311–37 333, 2021. DOI: 10.1109/ACCESS.2021.3061451.

9 K. Naveed, F. Abdullah, **H. A. Madni**, M. A. Khan, T. M. Khan, and S. S. Naqvi, "Towards Automated Eye Diagnosis: An Improved Retinal Vessel Segmentation Framework Using Ensemble Block Matching 3D Filter," *Diagnostics*, vol. 11, no. 1, 2021, ISSN: 2075-4418. DOI: 10.3390/diagnostics11010114.

10 M. Raza, K. Naveed, A. Akram, *et al.*, "DAVS-NET: Dense Aggregation Vessel Segmentation Network for retinal vasculature detection in fundus images," *Plos one*, vol. 16, no. 12, e0261698, 2021. DOI: 10.1371/journal.pone.0261698.

11 M. Tabassum, T. M. Khan, M. Arsalan, *et al.*, "CDED-Net: Joint Segmentation of Optic Disc and Optic Cup for Glaucoma Screening," *IEEE Access*, vol. 8, pp. 102 733–102 747, 2020. DOI: 10.1109/ACCESS.2020.2998635.

Conference Proceedings

1 **H. A. Madni**, R. M. Umer, and G. L. Foresti, "Federated Learning for Data and Model Heterogeneity in Medical Imaging," in *International Conference on Image Analysis and Processing*, Springer, 2023, pp. 167–178. DOI: 10.1007/978-3-031-51026-7_15.

2 A. Ahmed, A. Manzoor, A. Khan, *et al.*, "Performance measurement of energy management controller using heuristic techniques," in *Complex, Intelligent, and Software Intensive Systems: Proceedings of the 11th International Conference on Complex, Intelligent, and Software Intensive Systems (CISIS-2017)*, Springer, 2018, pp. 181–188. DOI: 10.1007/978-3-319-61566-0_17.

3 **H. A. Madni**, Z. Anwar, and M. A. Shah, "Data mining techniques and applications — A decade review," in *2017 23rd International Conference on Automation and Computing (ICAC)*, 2017, pp. 1–7. DOI: 10.23919/IConAC.2017.8082090.

Skills

Languages	English, Urdu.
Coding	Python, Java, C++, PHP, SQL, …
Databases	Oracle, MySQL, PostgreSQL.
Web Dev	HTML, CSS, JavaScript, Apache Web Server, Tomcat Web Server.
Misc.	Academic research, teaching, training, LaTeX typesetting and publishing.

Certification

2024 **Oxford Machine Learning Summer School**, Oxford, UK.

2023 **International Computer Vision Summer School (ICVSS2023)**, Sicily, Italy.

2022 **International Summer School on Artificial Intelligence**, University of Udine, Italy.

2014 **Professional Software Development**, Aptech Education, Pakistan.

Awards and Achievements

2021 **MIUR, Italy Scholarship** for PhD in Computer Science and Artificial Intelligence.

Awards and Achievements (continued)

2017 **PEEF, Pakistan Scholarship** for a Master's in Computer Science.

2011 **Merit Scholarship**, ICT R&D Fund, Pakistan for Bachelor in Computer System Engineering.

References

Prof. Dr. Carsten Marr
Professor at Institute of AI for Health
Helmholtz Munich,
Munich, Germany.
carsten.marr@helmholtz-munich.de

Dr. Rao Muhammad Umer
PostDoc Researcher at Institute of AI for Health,
Helmholtz Munich,
Munich, Germany.
umer.rao@helmholtz-munich.de